

Chapter 34

Defining the Specificity of Blood-Group-Binding Lectins Through Microarray Analysis

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Abstract

The assessment of blood group antigens is critical to transfusion medicine practice, as accurately identifying these antigens in both donors and patients is key to ensuring safe transfusions. Although plant lectins have long been used to aid in the identification of many carbohydrate-based blood group antigens, recent research has shown that a family of mammalian proteins known as galectins can also recognize and bind to blood group antigens. However, determining the fine specificity of lectins toward these antigens is challenging because of the structural complexity of the antigens and past limitations in the resources available to characterize their interactions in detail. To overcome this, the Consortium for Functional Glycomics (CFG) developed glycan microarrays composed of chemoenzymatically defined glycans, including a wide range of blood group antigens. This platform enables precise analysis of lectin–antigen interactions and uses additional glycans as controls to refine the specificity of these interactions. We will describe the use of this platform to characterize the specificity and relative affinity of lectins for blood group and related glycans.

Key words CFG glycan array, Galectins, Blood group antigens, Self-like antigens

1 Introduction

Prior to the characterization of the ABO(H) blood group system, the outcomes of blood transfusion were unpredictable [1]. Initial attempts at transfusion yielded inconsistent results; while some recipients tolerated transfusion without apparent complication, others experienced severe, and sometimes fatal, adverse events [1–4]. The seminal discovery by Karl Landsteiner in 1900 of the ABO blood groups and their corresponding antibodies not only represented the first documented example of human polymorphism but also established a mechanistic framework for understanding transfusion incompatibility and predicting transfusion outcomes

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[5–7]. As a result, routine testing for ABO antigens and their specific antibodies has become standard practice for all transfusion recipients, rendering these the most frequently assessed antigens and polymorphisms in the human population [8–10].

Anti-ABO antibodies typically emerge within the first year of life, independent of prior transfusion or pregnancy, and act as the most common immunological barrier to transfusion and transplantation [11, 12]. Although the precise mechanisms underlying their formation remain incompletely understood, environmental exposure and host-related factors have been proposed as possible contributors to this process [11, 13–15]. The generation of naturally occurring antibodies against ABO and related glycan antigens differs from that of other red blood cell (RBC) alloantibodies, which generally arise only after antigenic exposure through allogeneic transfusion or pregnancy [16–21]. While much remains to be learned about the pathways that govern both naturally arising and exposure-induced alloantibody formation, the establishment of preclinical models for studying alloimmunization in general has driven significant progress in the field, as illustrated by numerous models and overall studies that have begun to elucidate key factors that drive this process [21–31]. Findings from patient studies and these experimental systems have begun to clarify critical immune players and pathways involved, with each study revealing distinct features of this process following exposure to unique alloantigens that collectively suggest that this process is more complicated than previously appreciated [32–40]. Moreover, factors that influence the outcomes of incompatible transfusion, such as antibody engagement, RBC clearance, and the overall consequences of hemolysis, are now being increasingly understood through a similar use of a variety of model systems and parallel clinical studies [41–46].

Despite being recognized for more than a century, the fine structural characteristics of ABO and H antigens remain incompletely defined [3–6]. While the core features of A, B, and H epitopes have long been established, these determinants reside on structurally complex glycan scaffolds attached to glycolipids and glycoproteins of the RBC membrane. Subtle variation in presentation, linkage, and context can modulate the way antibodies and other carbohydrate-binding proteins recognize these structures. This complexity bears direct clinical relevance, as in the distinction between A1 and A2 subgroups, a distinction that has implications for transfusion and transplantation compatibility [47]. Traditional diagnostic tools such as plant-derived lectins can assist in resolving A1 versus A2 [48], but studies have shown that these reagents are imperfect. In the transplant setting, for example, lectin-based discrimination may underestimate compatibility, excluding potential donor–recipient matches [47, 49].

Beyond exogenous plant lectins, mammals produce their own families of lectins with physiologic roles in glycan recognition, including blood group and related antigens [50]. Among these are the galectins, a broadly conserved family of β-galactoside-binding proteins. Each galectin harbors a carbohydrate-recognition domain (CRD) formed by two antiparallel β-sheets folded into a β-sandwich, generating the canonical galectin-binding pocket. Within this CRD, key residues, including histidine, arginine, asparagine, tryptophan, and glutamic acid, form a conserved "carbohydrate-binding cassette" that facilitate recognition of β-galactose moieties [51–55]. Sequence variations outside this conserved cassette are thought to diversify binding specificity across the galectin family and confer differential recognition of more complex glycans, such as the ABO(H) and Lewis antigens [56].

Defining the fine specificity of lectin–glycan interactions has historically been challenging. Early studies inferred binding patterns largely from hemagglutination assays, exploiting the capacity of lectins to crosslink and therefore agglutinate RBCs [57–59]. While useful, such assays are unable to resolve which specific glycan moieties underlie high-affinity recognition, given that the RBC surface displays a heterogeneous and densely glycosylated landscape. To overcome these challenges, high-throughput glycan microarray platforms, such as those developed by the Consortium for Functional Glycomics (CFG), have been applied to systematically assess galectin–glycan interactions, which have provided important insight into the binding specificity of distinct family members and their overall function [50, 60–67]. These arrays display hundreds of well-defined mammalian glycans, including ABO(H), α-Gal, N-acetyllactosamine, and Lewis antigens (A, B, X, Y), allowing direct comparison of binding preferences across multiple galectins [68].

Most prior analyses of galectin binding to glycan arrays have relied on single protein concentrations, which, although informative for outlining overall specificity, offer limited capacity to distinguish subtle differences in affinity. To address this limitation, we have reported an analysis approach of examining multiple human galectins across physiologically relevant concentrations (0.04–10 μM) on glycan arrays, as illustrated in a number of studies [60–62, 69, 70] (Fig. 1). This approach enables not only the accurate identification of preferred glycan ligands but also quantitative estimation of maximal binding (Bmax) and apparent dissociation constants (K_D) [69] (Fig. 2). The ability to resolve these parameters provides a more nuanced understanding of the fine specificity of galectins for blood group and related glycans, thereby advancing both the fundamental biology of lectin–glycan recognition and the development of refined reagents for transfusion and transplantation medicine [71].

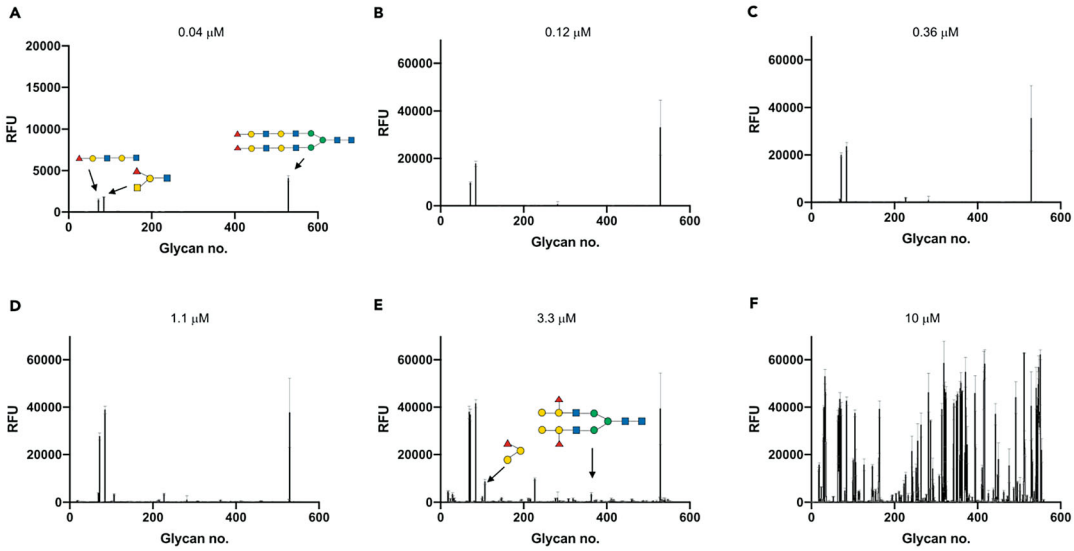


Fig. 1 Galectin-7 (Gal-7) exhibits specific interactions with blood group-containing structures across various concentrations. (**a**–**f**) Data from the Consortium for Functional Glycomics (CFG) glycan microarray, obtained after incubation at the following concentrations of Gal-7: 0.04 (**a**), 0.12 (**b**), 0.36 (**c**), 1.1 (**d**), 3.3 (**e**), and 10 (**f**) μM. RFU stands for relative fluorescence units. Error bars indicate the mean ± SD. (This research was originally published in iScience. Wu et al. [61]. © 2022 The Author(s))

2 Materials

2.1 Preparing Alexa Fluor™ 488 Reagent for Coupling

1. Phosphate-buffered saline (PBS) 1X.
2. Alexa Fluor™ 488 carboxylic acid, succinimidyl ester (Invitrogen) by Thermo Fisher Scientific, A 20100.

2.2 Coupling Galectins with Alexa Fluor 488 for CFG Arrays

1. Recombinantly purified full-length human galectins are stored in an elution buffer containing 14 mM β-mercaptoethanol and 100 mM lactose.
2. PD-10 desalting column, Sephadex™ G-25 M, GE, 17085101.
3. Lactose–Sepharose column for preparing active galectins for glycan array analysis.
4. Lactose.
5. Spectrophotometer.
6. Cuvettes for spectrophotometer.
7. 1.7-mL microcentrifuge tubes.
8. Amicon centrifugal filters.
9. Centrifuge.

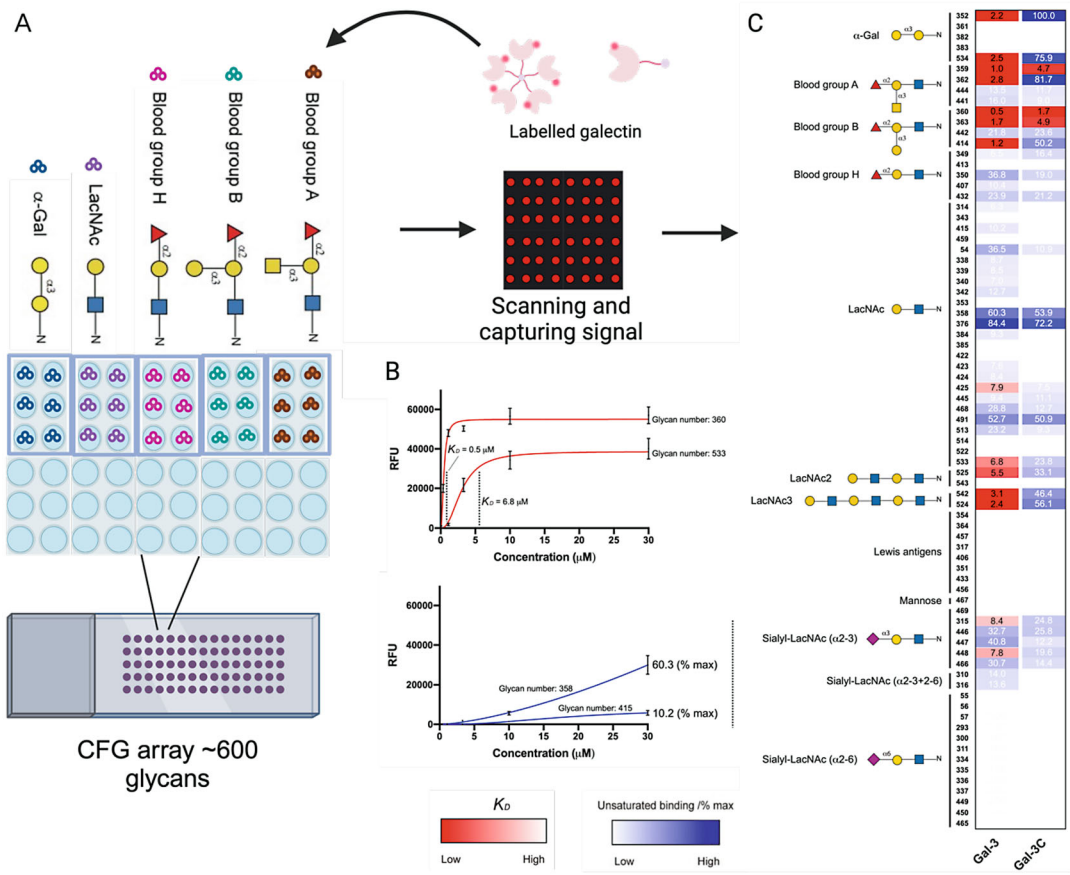


Fig. 2 K_D values for the binding of galectin-3 (Gal-3) and galectin-3C (Gal-3C) to blood group antigens and other mammalian glycans. (**a**) Illustration of the process of using the CFG array. (**b**) Representative binding isotherms were used to determine K_D values and the % maximum binding for unsaturated glycans. (**c**) Blood group antigens are shown with heat maps indicating K_D values (in red) and % maximum binding (in blue). (This research was originally published in *Frontiers in Microbiology* with some modifications. Wu et al.. © 2021 Wu, Ho, Kamili, Wang, Murdock, Cummings, Arthur and Stowell)

2.3 Method for Performing a CFG Glycan Microarray

1. Mammalian glycan array slides provided by the Consortium for Functional Glycomics (CFG).
2. Forceps.
3. Coplin jars.
4. Coverslips to cover the slides after loading galectins.
5. End-over-end rotator shaker.
6. Orbital shaker platform.
7. Blocking buffer: 1% bovine serum albumin (BSA) in 1X PBS with 0.05% Tween-20.
8. Vacuum desiccator.

9. Molecular Devices GenePix 4000B Microarray Scanner.
10. GenePix Pro 7 Software for imaging the slides and analyzing the data.

3 Methods

3.1 Preparing Alexa Fluor™ 488 Reagent for Coupling

1. Add 500 μL of 1X PBS to 5 mg of Alexa Fluor 488.
2. Aliquot into 5 tubes, with 100 μL in each tube, and store at −20 °C.

3.2 Coupling Galectins with Alexa Fluor 488 for CFG Arrays

1. Desalt the recombinant galectins using a PD-10 desalting column to remove 2-mercaptoethanol and lactose. Use cold 1X PBS for wash and elution.
2. Check the absorbance of all eluted fractions at OD 280 nm.
3. Combine the OD 280-nm positive fractions and concentrate. The concentration of the protein used for coupling should be at least 2 mg/mL for optimal results.
4. Add lactose at a 100 mM concentration (36 mg/mL).
5. Add 50 μL of Alexa Fluor 488 dye (10 mg/mL) to 1 mL of galectin (2 mg/mL) solution (*see* **Note 1**).
6. Incubate at room temperature for 1 h on a rotator shaker, avoiding light.
7. After coupling, pass the Alexa Fluor 488-coupled galectin through a PD-10 desalting column to remove any unbound dye and lactose.
8. The collected protein fraction is pooled and passed through 2 mL of lactose–Sepharose column, followed by washing the column with PBS.
9. The bound protein is eluted using 100 mM lactose-containing elution buffer.
10. This fraction contains only the active galectins. Pass this active galectin fraction through a PD-10 desalting column again to remove the lactose.
11. Use the manufacturer's instructions for calculating protein concentration post-labeling.

3.3 Method for Performing a CFG Glycan Microarray

1. Take the slides from the −20 °C freezer and place them in a Coplin jar using forceps.
2. Place the Coplin jar inside the vacuum desiccator. Connect the vacuum and close the valve. Leave the slides in the vacuum for 30 minutes (*see* **Note 2**).

3. Block the slides in blocking buffer containing 1% bovine serum albumin (BSA) in PBST (1X PBS with 0. 05% Tween-20).
4. Prepare PBST, PBS, and distilled water in Coplin jars.
5. Rehydrate the CFG array slide in PBST for 5 minutes.
6. Dilute galectin in blocking buffer to different concentrations: 0.04, 0.12, 0.36, 1.1, 3.3, and 10 μM.
7. Add 200 μL of galectin solution to the slide, and place the microscope cover glass over it.
8. Incubate for 1 h at room temperature, avoiding light.
9. Wash with PBST (4 times).
10. Wash with 1X PBS (4 times).
11. Wash with water (4 times).
12. Spin-dry.
13. Scan the image using GenePix Microarray Scanner at 523 nm (*see* **Note 3**). Save the image as a Tiff file.
14. After acquiring the image, it is aligned to an array map called GenePix Array list file or.gal file that indicates the coordinates of the spots by numbers correlated to the printed material [72].
15. The output file, or GenePix results file (.gpr), contains alignment information for analyzing the data using Microsoft Excel and can be further processed with Prism or GLAD (Glycan Array Dashboard) analysis tools (*see* **Note 4**).

4 Notes

1. Alexa Fluor 647 can also be used for labeling the galectins, followed by array analysis.
2. Exercise caution to ensure that the forceps do not come into contact with the glycan-printed areas, as any inadvertent touch could potentially compromise the integrity of the printed glycans and affect the accuracy of subsequent analyses or experiments.
3. Different scanner machines may have different PMT settings. Use of multiple concentrations of galectin can allow relative K_D values to be obtained.
4. GLAD (Glycan Array Dashboard) is a tool to visualize, analyze, and compare glycan microarray data developed by Dr. Akul Y. Mehta and Dr. Richard D. Cummings.

Conflicts of Interest

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